

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 01 • Proof

With Answers

Questions with full worked solutions

- 35 pages
- Past papers up to January 2026

PURE 2 | TOPIC 01

Proof

Your classified route

15

QUESTIONS

75

MARKS

4

FAMILIES

5

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 01 | FAMILY 01

Exhaustion/Cases

WHAT THIS FAMILY ASKS

Recognise the exhaustion/cases demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

V02 For modulo cases, use $3k, 3k+1, 3k+2$ or $4k, 4k+1, 4k+2, 4k+3$.

8 classified questions

36 total marks

T01

01_CASE

01_CASE_CORE

CHAPTER
ProofFAMILY
Exhaustion/casesVARIANT
core family move

5 marks

WMA12/01 JANUARY 2022

Question 10

Past-paper question

WMA12-01_2022_Jan_Q10

10. (i) Prove by counter example that the statement

“if p is a prime number then $2p + 1$ is also a prime number”

is not true.

(1)

(ii) Use proof by exhaustion to prove that if n is an integer then

$$5n^2 + n + 12$$

is always even.

(4)

(Total 5 marks)

T01 Proof

01_CASE Exhaustion/cases

01_CASE_CORE core family
move

WORKED SOLUTION

Question 10

WMA12/01 January 2022

WMA12-01_2022_Jan_Q10

1

Give a counterexampleTake $p = 7$, which is prime.

2

Show the statement fails $2p + 1 = 2(7) + 1 = 15$, and 15 is not prime. So the statement is false.

3

Case one: n evenIf n is even, let $n = 2k$. Then $5n^2 + n + 12 = 5(2k)^2 + 2k + 12 = 20k^2 + 2k + 12 = 2(10k^2 + k + 6)$, which is even.

4

Case two: n oddIf n is odd, let $n = 2k + 1$. Then $5n^2 + n + 12 = 5(2k + 1)^2 + (2k + 1) + 12$.

5

Simplify the odd case $5(2k + 1)^2 + (2k + 1) + 12 = 20k^2 + 22k + 18 = 2(10k^2 + 11k + 9)$, which is even.

6

Conclude by exhaustionEvery integer is either even or odd, and both cases give an even result. Therefore $5n^2 + n + 12$ is always even.**FINAL ANSWER**

WMA12-01_2022_Jan_Q10

(i) For $p = 7$, $2p + 1 = 15$, not prime. (ii) Proven.

T01

01_CASE

01_CASE_CORE

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3 marks

WMA12/01 OCTOBER 2022

Question 1

Past-paper question

WMA12-01_2022_Oct_Q01

1. Given that a , b and c are integers greater than 0 such that

- $c = b + 2$
- $a + b + c = 10$

Prove, by exhaustion, that the product of a , b and c is always even.

You may use the table below to illustrate your answer.

(3)

You may not need to use all rows of this table.

a	b	c
	1	
	2	

T01 Proof

01_CASE Exhaustion/cases

01_CASE_CORE core family
move

WORKED SOLUTION

Question 1

WMA12/01 October 2022

WMA12-01_2022_Oct_Q01

1

Use the two conditions

Since $c = b + 2$ and $a + b + c = 10$, substitute for c : $a + b + (b + 2) = 10$.

2

Write a in terms of b

This gives $a + 2b + 2 = 10$, so $a = 8 - 2b$.

3

List the possible b values

The integers a, b, c are all positive. Since $b > 0$ and $a = 8 - 2b > 0$, the only possible values are $b = 1, 2, 3$.

4

Exhaust the cases

If $b = 1$, then $a = 6$ and $c = 3$, so $abc = 18$. If $b = 2$, then $a = 4$ and $c = 4$, so $abc = 32$. If $b = 3$, then $a = 2$ and $c = 5$, so $abc = 30$.

5

Conclude the proof

The list contains every possible case, and in every case the product is even. Therefore abc is always even.

FINAL ANSWER

WMA12-01_2022_Oct_Q01

The possible triples are $(6, 1, 3)$, $(4, 2, 4)$ and $(2, 3, 5)$, giving products 18, 32 and 30. Each product is even, so abc is always even.

T01

01_CASE

01_CASE_CORE

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5 marks

WMA12/01 MAY/JUNE 2023

Question 8

Past-paper question

WMA12-01_2023_MayJune_Q08

8. (i) A student writes the following statement:

“When a and b are consecutive **prime** numbers, $a^2 + b^2$ is never a multiple of 10”

Prove by counter example that this statement is **not** true.

(2)

- (ii) Given that x and y are even integers greater than 0 and less than 6, prove by exhaustion, that

$$1 < x^2 - \frac{xy}{4} < 15$$

(3)

T01 Proof

01_CASE Exhaustion/cases

01_CASE_CORE core family
move

WORKED SOLUTION

Question 8

WMA12/01 May/June 2023

WMA12-01_2023_MayJune_Q08

1

Choose consecutive primes

7 and 11 are consecutive prime numbers.

2

Check the statement $7^2 + 11^2 = 49 + 121 = 170$, which is a multiple of 10.

3

Conclude the counterexample

This single counterexample proves the student's statement is not true.

4

List the possible even valuesThe even integers greater than 0 and less than 6 are 2 and 4. Therefore the possible ordered pairs are $(2, 2)$, $(2, 4)$, $(4, 2)$, $(4, 4)$.

5

Check x equals twoIf $(x, y) = (2, 2)$, then $x^2 - \frac{xy}{4} = 4 - 1 = 3$. If $(x, y) = (2, 4)$, then $x^2 - \frac{xy}{4} = 4 - 2 = 2$.

6

Check x equals fourIf $(x, y) = (4, 2)$, then $x^2 - \frac{xy}{4} = 16 - 2 = 14$. If $(x, y) = (4, 4)$, then $x^2 - \frac{xy}{4} = 16 - 4 = 12$.

7

Complete the exhaustionThe possible values are 3, 2, 14, 12, and each lies strictly between 1 and 15. Hence $1 < x^2 - \frac{xy}{4} < 15$ in every possible case.**FINAL ANSWER**

WMA12-01_2023_MayJune_Q08

(i) A counterexample is $a = 7, b = 11$, since $7^2 + 11^2 = 170$. (ii) Exhaustion proves the inequality for $(x, y) = (2, 2), (2, 4), (4, 2), (4, 4)$.

T01

01_CASE

01_CASE_CORE

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core family move
3 marks

WMA12/01 OCTOBER 2023

Question 1

Past-paper question

WMA12-01_2023_Oct_Q01

1. Given that a , b and c are integers greater than 0 such that

- $c = 3a + 1$
- $a + b + c = 15$

prove, by exhaustion, that the product abc is always a multiple of 4

You may use the table below to illustrate your answer.

(3)

You may not need to use all rows of this table.

a	b	c	abc

T01 Proof

01_CASE Exhaustion/cases

01_CASE_CORE core family
move

WORKED SOLUTION

Question 1

WMA12/01 October 2023

WMA12-01_2023_Oct_Q01

1

Use the conditions

Since $c = 3a + 1$ and $a + b + c = 15$, substitute for c to get $a + b + 3a + 1 = 15$.

2

Write b in terms of a

This gives $b = 14 - 4a$.

3

List possible a values

The integers a, b, c are positive. Since $b = 14 - 4a > 0$, we need $a < 3.5$. Therefore $a = 1, 2, 3$.

4

Check each case

If $a = 1$, then $b = 10$ and $c = 4$, so $abc = 40$. If $a = 2$, then $b = 6$ and $c = 7$, so $abc = 84$. If $a = 3$, then $b = 2$ and $c = 10$, so $abc = 60$.

5

Conclude the exhaustion

The products 40, 84 and 60 are all multiples of 4. All possible cases have been checked, so abc is always a multiple of 4.

FINAL ANSWER

WMA12-01_2023_Oct_Q01

The only possible triples are $(1, 10, 4)$, $(2, 6, 7)$ and $(3, 2, 10)$, giving products 40, 84 and 60. Each is a multiple of 4, so abc is always a multiple of 4.

T01

01_CASE

01_CASE_CORE

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6 marks

WMA12/01 JANUARY 2024

Question 8

Past-paper question

WMA12-01_2024_Jan_Q08

8. (i) Use a counter example to show that the following statement is **false**

“ $n^2 + 3n + 1$ is prime for all $n \in \mathbb{N}$ ”

(2)

- (ii) Use algebra to prove by exhaustion that for all $n \in \mathbb{N}$

“ $n^2 - 2$ is **not** a multiple of 4”

(4)

T01 Proof

01_CASE Exhaustion/cases

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move

WORKED SOLUTION

Question 8

WMA12/01 January 2024

WMA12-01_2024_Jan_Q08

1

Choose a counterexampleTake $n = 6$.

2

Evaluate the expression

$$n^2 + 3n + 1 = 6^2 + 3(6) + 1 = 36 + 18 + 1 = 55.$$

3

Conclude false

55 is not prime, so the statement is false.

4

Use residues modulo fourEvery natural number is one of $4k$, $4k + 1$, $4k + 2$ or $4k + 3$.

5

Check n equals $4k$ If $n = 4k$, then $n^2 - 2 = 16k^2 - 2 = 4(4k^2 - 1) + 2$, not a multiple of 4.

6

Check n equals $4k$ plus oneIf $n = 4k + 1$, then $n^2 - 2 = 16k^2 + 8k - 1 = 4(4k^2 + 2k - 1) + 3$, not a multiple of 4.

7

Check n equals $4k$ plus twoIf $n = 4k + 2$, then $n^2 - 2 = 16k^2 + 16k + 2 = 4(4k^2 + 4k) + 2$, not a multiple of 4.

8

Check n equals $4k$ plus threeIf $n = 4k + 3$, then $n^2 - 2 = 16k^2 + 24k + 7 = 4(4k^2 + 6k + 1) + 3$, not a multiple of 4.

9

Complete the proofAll possible cases leave remainder 2 or 3 when divided by 4, so $n^2 - 2$ is not a multiple of 4.**FINAL ANSWER**

WMA12-01_2024_Jan_Q08

(i) **Counterexample:** $n = 6$ gives 55, which is not prime. (ii) **Proven by considering** $n = 4k, 4k + 1, 4k + 2, 4k + 3$.

T01

01_CASE

01_CASE_CORE

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6 marks

WMA12/01 OCTOBER 2024
Question 11

Past-paper question

WMA12-01_2024_Oct_Q11

11. (i) Prove by counter example that the statement

“If n is a prime number then $3^n + 2$ is also a prime number.”

is false.

(2)

(ii) Use proof by exhaustion to prove that if m is an integer that is **not** divisible by 3, then

$$m^2 - 1$$

is divisible by 3

(4)

T01 Proof

01_CASE Exhaustion/cases

01_CASE_CORE core family
move

WORKED SOLUTION

Question 11

WMA12/01 October 2024

WMA12-01_2024_Oct_Q11

1

Choose a prime valueTake $n = 5$, which is prime.

2

Evaluate the expression $3^5 + 2 = 243 + 2 = 245 = 5 \times 49$, so it is not prime. This disproves the statement.

3

List the possible forms of m If integer m is not divisible by 3, then m has remainder 1 or 2 on division by 3.

4

Case oneIf $m = 3q + 1$, then $m^2 - 1 = (3q + 1)^2 - 1 = 9q^2 + 6q = 3(3q^2 + 2q)$, which is divisible by 3.

5

Case twoIf $m = 3q + 2$, then $m^2 - 1 = (3q + 2)^2 - 1 = 9q^2 + 12q + 3 = 3(3q^2 + 4q + 1)$, which is divisible by 3.

6

Conclude by exhaustionBoth possible cases for an integer not divisible by 3 have been checked, so $m^2 - 1$ is divisible by 3.**FINAL ANSWER**

WMA12-01_2024_Oct_Q11

(i) **Counterexample:** $n = 5$ gives $3^5 + 2 = 245$, not prime. (ii) **Proven.**

T01

01_CASE

01_CASE_CORE

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4 marks

WMA12/01 MAY/JUNE 2025

Question 10

Past-paper question

WMA12-01_2025_MayJune_Q10

10: In this question you must show detailed reasoning.

Use algebra to prove by exhaustion that,

for all positive integers m that are **not** multiples of 3, the value of

$$m^2 + 3m + 2$$

is always a multiple of 3

(4)

T01 Proof

01_CASE Exhaustion/cases

01_CASE_CORE core family
move

WORKED SOLUTION

Question 10

WMA12/01 May/June 2025

WMA12-01_2025_MayJune_Q10

1

List the possible cases

A positive integer that is not a multiple of 3 must have the form $m = 3q + 1$ or $m = 3q + 2$.

2

Case one

If $m = 3q + 1$, then $m^2 + 3m + 2 = (3q + 1)^2 + 3(3q + 1) + 2 = 9q^2 + 15q + 6 = 3(3q^2 + 5q + 2)$.

3

Case two

If $m = 3q + 2$, then $m^2 + 3m + 2 = (3q + 2)^2 + 3(3q + 2) + 2 = 9q^2 + 21q + 12 = 3(3q^2 + 7q + 4)$.

4

Conclude by exhaustion

Both possible cases give a factor of 3, so $m^2 + 3m + 2$ is always a multiple of 3.

FINAL ANSWER**Proven.**

WMA12-01_2025_MayJune_Q10

T01

01_CASE

01_CASE_HV01

CHAPTER
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VARIANT
For modulo cases, use
/(3k,3k+1,3k+2/) or
/(4k,4k+1,4k+2,4k+3/).
4 marks

WMA12/01 JANUARY 2023
Question 10

Past-paper question

WMA12-01_2023_Jan_Q10

10. A student was asked to prove by exhaustion that

if n is an integer then $2n^2 + n + 1$ is **not** divisible by 3

The start of the student's proof is shown in the box below.

Consider the case when $n = 3k$

$$2n^2 + n + 1 = 18k^2 + 3k + 1 = 3(6k^2 + k) + 1$$

which is not divisible by 3

Complete this proof.

(4)

T01 Proof

01_CASE Exhaustion/cases

01_CASE_HV01 For modulo cases, use $\wedge(3k, 3k+1, 3k+2)$ or $\wedge(4k, 4k+1, 4k+2, 4k+3)$.

WORKED SOLUTION

Question 10

WMA12/01 January 2023

WMA12-01_2023_Jan_Q10

1

Use the three possible forms

Every integer is exactly one of the forms $3k$, $3k + 1$ or $3k + 2$, where k is an integer.

2

First case already given

For $n = 3k$, the expression becomes $3(6k^2 + k) + 1$, so it leaves remainder 1 when divided by 3.

3

Try n equals $3k$ plus 1

If $n = 3k + 1$, then $2n^2 + n + 1 = 2(3k + 1)^2 + (3k + 1) + 1$.

4

Simplify the second case

$2(9k^2 + 6k + 1) + 3k + 2 = 18k^2 + 15k + 4 = 3(6k^2 + 5k + 1) + 1$, which is not divisible by 3.

5

Try n equals $3k$ plus 2

If $n = 3k + 2$, then $2n^2 + n + 1 = 2(3k + 2)^2 + (3k + 2) + 1$.

6

Simplify the third case

$2(9k^2 + 12k + 4) + 3k + 3 = 18k^2 + 27k + 11 = 3(6k^2 + 9k + 3) + 2$, which is not divisible by 3.

7

Complete the exhaustion

All possible integer forms have been checked, and none gives a multiple of 3. Therefore $2n^2 + n + 1$ is not divisible by 3.

FINAL ANSWER

WMA12-01_2023_Jan_Q10

For $n = 3k$, $n = 3k + 1$ and $n = 3k + 2$, the expression has remainders 1, 1 and 2 respectively when divided by 3. Therefore $2n^2 + n + 1$ is never divisible by 3.

TOPIC 01 | FAMILY 02

Disproof

WHAT THIS FAMILY ASKS

Recognise the disproof demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

2 classified questions

13 total marks

T01

01_COUNTER

01_COUNTER_CORE

CHAPTER

Proof

FAMILY

Disproof

VARIANT

core family move

7 marks

WMA12/01 MAY/JUNE 2022

Question 3

Past-paper question

WMA12-01_2022_MayJune_Q03

3. (i) Show that the following statement is **false**:

“($n + 1$)³ – n^3 is prime for all $n \in \mathbb{N}$ ”

(2)

- (ii) Given that the points $A(1, 0)$, $B(3, -10)$ and $C(7, -6)$ lie on a circle, prove that AB is a diameter of this circle.

(5)

T01 Proof

01_COUNTER Disproof

01_COUNTER_CORE core
family move

WORKED SOLUTION

Question 3

WMA12/01 May/June 2022

WMA12-01_2022_MayJune_Q03

1

Choose a counterexampleTake $n = 5$.

2

Evaluate the expression

$$(5 + 1)^3 - 5^3 = 216 - 125 = 91.$$

3

Show it is not prime $91 = 7 \times 13$, so the statement is false.

4

Find the gradient CAFor $A(1, 0)$ and $C(7, -6)$, the gradient of CA is $\frac{0 - (-6)}{1 - 7} = -1$.

5

Find the gradient CBFor $B(3, -10)$ and $C(7, -6)$, the gradient of CB is $\frac{-10 - (-6)}{3 - 7} = 1$.

6

Show the right angleThe product of the gradients is -1 , so CA is perpendicular to CB and $\angle ACB = 90^\circ$.

7

Conclude the diameterAn angle of 90° at the circumference subtends a diameter, so AB is a diameter of the circle.**FINAL ANSWER**

WMA12-01_2022_MayJune_Q03

(i) Counterexample: $n = 5$ gives 91, not prime. (ii) Proven: AB is a diameter.

T01

01_COUNTER

01_COUNTER_CORE

CHAPTER

Proof

FAMILY

Disproof

VARIANT

core family move

6 marks

WMA12/01 MAY/JUNE 2024

Question 5

Past-paper question

WMA12-01_2024_MayJune_Q05

5. In this question you must show detailed reasoning.

- (a) Given that x and y are positive numbers such that

$$(x - y)^3 > x^3 - y^3$$

prove that

$$y > x$$

(4)

- (b) Using a counter example, show that the result in part (a) is not true for all real numbers.

(2)

T01 Proof

01_COUNTER Disproof

01_COUNTER_CORE core
family move

WORKED SOLUTION

Question 5

WMA12/01 May/June 2024

WMA12-01_2024_MayJune_Q05

1

Start from the inequality

$$(x - y)^3 > x^3 - y^3.$$

2

Expand the left side

$$x^3 - 3x^2y + 3xy^2 - y^3 > x^3 - y^3.$$

3

Cancel common terms

Subtract $x^3 - y^3$ from both sides to get $-3x^2y + 3xy^2 > 0$.

4

Factor

$$3xy(y - x) > 0.$$

5

Use positivity

Since x and y are positive, $3xy > 0$. Therefore $y - x > 0$, so $y > x$.

6

Give a real-number counterexample

Take $x = 1$ and $y = -1$. Then $(x - y)^3 = 2^3 = 8$ and $x^3 - y^3 = 1 - (-1) = 2$, so the given inequality is true.

7

Show the conclusion fails

But $y > x$ would mean $-1 > 1$, which is false. Therefore the result is not true for all real numbers.

FINAL ANSWER

WMA12-01_2024_MayJune_Q05

(a) **Proven.** (b) **For example, $x = 1, y = -1$ satisfies the inequality but not $y > x$.**

TOPIC 01 | FAMILY 03

Algebraic Divisibility

WHAT THIS FAMILY ASKS

Recognise the algebraic divisibility demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**3 classified questions**

18 total marks

T01

01_FACTOR

01_FACTOR_CORE

CHAPTER

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FAMILY

Algebraic divisibility

VARIANT

core family move

5 marks

WMA12/01 MAY/JUNE 2021

Question 3

Past-paper question

WMA12-01_2021_MayJune_Q03

3. (i) Prove that for all single digit prime numbers, p ,

$$p^3 + p \text{ is a multiple of } 10$$

(2)

- (ii) Show, using algebra, that for $n \in \mathbb{N}$

$$(n + 1)^3 - n^3 \text{ is not a multiple of } 3$$

(3)

T01 Proof

01_FACTOR Algebraic
divisibility01_FACTOR_CORE core
family move

WORKED SOLUTION

Question 3

WMA12/01 May/June 2021

WMA12-01_2021_MayJune_Q03

1

List the cases

The single digit prime numbers are 2, 3, 5, and 7.

2

Check each prime $2^3 + 2 = 10$, $3^3 + 3 = 30$, $5^3 + 5 = 130$, and $7^3 + 7 = 350$.

3

Conclude part oneEach result is a multiple of 10, so $p^3 + p$ is a multiple of 10 for all single digit primes.

4

Expand the difference of cubes $(n + 1)^3 - n^3 = (n^3 + 3n^2 + 3n + 1) - n^3 = 3n^2 + 3n + 1$.

5

Factor the multiple of three $3n^2 + 3n + 1 = 3n(n + 1) + 1$.

6

Use the remainder $3n(n + 1)$ is a multiple of 3, so $3n(n + 1) + 1$ is one more than a multiple of 3. Therefore it is not a multiple of 3.**FINAL ANSWER**

WMA12-01_2021_MayJune_Q03

(i) **Proven by checking** $p = 2, 3, 5, 7$. (ii) **Proven:** $(n + 1)^3 - n^3$ **leaves remainder 1 on division by 3.**

T01

01_FACTOR

01_FACTOR_CORE

CHAPTER

Proof

FAMILY

Algebraic divisibility

VARIANT

core family move

5 marks

WMA12/01 JANUARY 2025

Question 8

Past-paper question

WMA12-01_2025_Jan_Q08

8. (i) A student states

“If x and y are irrational numbers, $x \neq y$, then xy is also irrational.”

Show, by counter example, that this statement is not always true.

(1)

- (ii) Prove, using algebra, that for all odd integers n , the value of the expression

$$n^3 + 3n + 2$$

is always even but never a multiple of 4

(4)

T01 Proof

01_FACTOR Algebraic
divisibility01_FACTOR_CORE core
family move

WORKED SOLUTION

Question 8

WMA12/01 January 2025

WMA12-01_2025_Jan_Q08

1

Give a counterexample

Let $x = \sqrt{2}$ and $y = \frac{1}{\sqrt{2}}$. Both are irrational and $x \neq y$, but $xy = 1$, which is rational.

2

Represent an odd integer

Let $n = 2q + 1$, where q is an integer.

3

Substitute into the expression

$$n^3 + 3n + 2 = (2q + 1)^3 + 3(2q + 1) + 2.$$

4

Expand and factor

$$n^3 + 3n + 2 = 8q^3 + 12q^2 + 12q + 6 = 2(4q^3 + 6q^2 + 6q + 3).$$

5

Show it is even

The expression has a factor of 2, so it is even.

6

Show it is not a multiple of four

The bracket $4q^3 + 6q^2 + 6q + 3$ is odd, because the first three terms are even and 3 is odd. Therefore the expression is 2 times an odd integer, so it is not divisible by 4.

FINAL ANSWER

WMA12-01_2025_Jan_Q08

(i) For example, $x = \sqrt{2}$ and $y = \frac{1}{\sqrt{2}}$ give $xy = 1$. (ii) Proven.

T01

01_FACTOR

01_FACTOR_CORE

CHAPTER

Proof

FAMILY

Algebraic divisibility

VARIANT

core family move

8 marks

WMA12/01 JANUARY 2026

Question 6

Past-paper question

WMA12-01_2026_Jan_Q06

6. (i) Given that p and q are consecutive odd numbers, where $p > q > 0$, prove that

$$p^2 - q^2$$

is a multiple of 8

(4)

- (ii) The curve C has equation

$$y = x^3 + 12x^2 + 49x + 2$$

Prove that C has no stationary points.

(4)

T01 Proof

01_FACTOR Algebraic
divisibility01_FACTOR_CORE core
family move

WORKED SOLUTION

Question 6

WMA12/01 January 2026

WMA12-01_2026_Jan_Q06

1

Represent the consecutive odd numbers

Let $q = 2n - 1$ and $p = 2n + 1$, where n is a positive integer. Then $p > q > 0$ and the numbers are consecutive odd numbers.

2

Subtract the squares

$$p^2 - q^2 = (2n + 1)^2 - (2n - 1)^2.$$

3

Simplify the expression

$$p^2 - q^2 = (4n^2 + 4n + 1) - (4n^2 - 4n + 1) = 8n.$$

4

Conclude the proof

Since n is an integer, $8n$ is a multiple of 8, so $p^2 - q^2$ is a multiple of 8.

5

Differentiate the curve

For $y = x^3 + 12x^2 + 49x + 2$, $\frac{dy}{dx} = 3x^2 + 24x + 49$.

6

Show the gradient is never zero

$3x^2 + 24x + 49 = 3(x + 4)^2 + 1$. This is always positive for real x .

7

State the conclusion

Since $\frac{dy}{dx}$ is never 0, the curve has no stationary points.

FINAL ANSWER

WMA12-01_2026_Jan_Q06

(i) Proven. (ii) C has no stationary points.

TOPIC 01 | FAMILY 04

Inequality Proof

WHAT THIS FAMILY ASKS

Recognise the inequality proof demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

2 classified questions

8 total marks

T01

01_INEQ

01_INEQ_CORE

CHAPTER
Proof

FAMILY
Inequality proof

VARIANT
core family move

4 marks

WMA12/01 JANUARY 2021

Question 5

Past-paper question

WMA12-01_2021_Jan_Q05

5. (i) Use algebra to prove that for all $x \geq 0$

$$3x + 1 \geq 2\sqrt{3x}$$

(3)

- (ii) Show that the following statement is not true.

“The sum of three consecutive prime numbers is always a multiple of 5”

(1)

T01 Proof

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move

WORKED SOLUTION

Question 5

WMA12/01 January 2021

WMA12-01_2021_Jan_Q05

1

Start from a square

For $x \geq 0$, $\sqrt{3x}$ is real and $(\sqrt{3x} - 1)^2 \geq 0$.

2

Expand the square
$$(\sqrt{3x} - 1)^2 = 3x - 2\sqrt{3x} + 1 \geq 0.$$

3

Rearrange the inequality

Therefore $3x + 1 \geq 2\sqrt{3x}$, as required.

4

Give a counterexample

The consecutive prime numbers 5, 7, and 11 have sum $5 + 7 + 11 = 23$.

5

Complete the disproof

23 is not a multiple of 5, so the statement is not true.

FINAL ANSWER

WMA12-01_2021_Jan_Q05

(i) **Proven.** (ii) **For example, $5 + 7 + 11 = 23$, which is not a multiple of 5.**

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WMA12/01 OCTOBER 2021

Question 9

Past-paper question

WMA12-01_2021_Oct_Q09

9. (a) Prove that for all positive values of x and y ,

$$\frac{x+y}{2} \geq \sqrt{xy} \quad (3)$$

- (b) Prove by counter-example that this inequality does not hold when x and y are both negative.

(1)

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WORKED SOLUTION

Question 9

WMA12/01 October 2021

WMA12-01_2021_Oct_Q09

1

Start with a square

For positive x and y , both $\sqrt{x}$ and $\sqrt{y}$ are real, so $(\sqrt{x} - \sqrt{y})^2 \geq 0$.

2

Expand

$$(\sqrt{x} - \sqrt{y})^2 = x - 2\sqrt{xy} + y \geq 0.$$

3

Rearrange

Thus $x + y \geq 2\sqrt{xy}$, and dividing by 2 gives $\frac{x+y}{2} \geq \sqrt{xy}$.

4

Give a negative counterexample

Take $x = -1$ and $y = -1$. Then $\frac{x+y}{2} = -1$, while $\sqrt{xy} = \sqrt{1} = 1$.

5

Conclude

Since $-1 \geq 1$ is false, the inequality does not hold for all negative x and y .

FINAL ANSWER

WMA12-01_2021_Oct_Q09

(a) **Proven.** (b) **For example, $x = y = -1$ gives $-1 \not\geq 1$.**